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Software Development Life Cycle and Agile Principles

Assignment - 1

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is an organised method used by software engineers and developers to build reliable, efficient and maintainable systems. It gurantees thats software is provided in a structured, cost-effective and timely manner. Each phase builds upon the previous one, resulting a logical flow from the beginning of a project to its end.

Phases of SDLC

i. Requirements Gathering: This is where everything begins. Understanding what the client needs, what problems the System should solve, and the features it must include.
Importance: clear, detailed requirements assist to avoid costly changes later and set the groundwork for the entire project.
Key outputs: Requirement Specifications Document and Use Case Scenarios.

ii. System Design: Creating a blueprint for building the System based on obtained requirements. This comprises architecture, user interface design, data flow, and technology Stack selection.
Importance: A well-thought-out design provides a straight-forward execution while minimising future technological debt.
Key outputs: Design Diagrams (UML, ER), Technical Architecture Document.

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iii. Implementation (coding): Developers are now writing code in accordance with design documents and best development standards.

Importance: This is where abstraction meets reality. The proper implementation influences performance, scalability and maintainability.

Key Focus: Code reviews, version control, and modular development

iv. Testing: Conduct rigorous tests to identify bugs, confirm functionallity, and ensure quality. Testing can be done manually or automatically.

Importance: Testing safeguards consumers against a negative experience and gurantees that the program performs as intended in various environments.

Types: Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).

V. Deployment and Maintenance: The program is made available to a limited audience (beta) or the general audience. post-deployment upgrades and bug fixes are performed as necessary.

Importance: Successful deployment indicates that the product is ready to offer value. Ongoing maintenance ensures thpt it remains relevant and functional.

Includes: Deployment Plans, Feedback loops, Continuous Improvement.

Connection between them:

- Each phase leads into the next, resulting in a logical development.
- Mistakes in the early stages (such as unclear requirements) have an impact on subsequent phases.

- Feedback loops are common, particularly between testing and implementation or deployment and requirements (updates).

The SDLC is a framework for ensuring software is well-planned, carefully produced and properly tested. Understanding the objective of each step and its part in the larger picture allows for the development of better software that actually serves its consumers.